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Paper Code : IT-504D

OPERATIONS RESEARCH

Time Allotted : 3 Hours | *Full Marks* : 70

The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable.

GROUP – A

(Multiple Choice Type Questions)

1. Choose the correct alternatives for any *ten* of the following : $10 \times 1 = 10$

   i) A feasible solution to an LP problem

   a) must satisfy all of the problems constraints simultaneously

   b) need not satisfy all of the constraints, only some of them

   c) must be corner point of the feasible region

   d) must optimize the value of the objective function.

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ii) For a maximization problem, the objective function co-efficient for an artificial variable is

a) $+M$ b) $-M$

c) zero d) none of these.

iii) Operations Research deals with problems associated with

a) Decision making b) Scarce resources

c) Model building d) All of these.

iv) A necessary and sufficient condition for a basic solution of a minimization type problem to be optimal is that (for all j)

a) $Z_j - C_j \geq 0$

b) $Z_j - C_j \leq 0$

c) $Z_j - C_j = 0$

d) $Z_j - C_j > 0$ *or* $Z_j - C_j < 0$.

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v) A transportation problem is a balanced transportation problem if

a) Total demand and total supply are equal and number of sources equals to the number of destinations

b) Total demand equals to the total supply irrespective of the number of sources and destinations

c) Number of sources matches with the number of destinations

d) The corresponding basic feasible solution is to be degenerate.

vi) In an assignment problem the minimum number of lines covering all zeros in the reduced cost matrix of order n can be

a) at most n b) $n + 1$

c) $n - 1$ d) at least n.

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vii) The steady state behaviour of the Queuing system which is in operation for a sufficient long period of time exists if

a) $\lambda > \mu$ b) $\lambda < \mu$

c) $\lambda = \mu$ d) $\lambda \leq \mu$.

viii) General queuing models may be completely specified by the symbol (a/b/c) : (d/e) is known as

a) Thompson's notation

b) Poisson's notation

c) Kendall's notation

d) None of these.

ix) The full form of CPM related with network scheduling is

a) compressed path method

b) critical position method

c) critical point method

d) critical path method.

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x) Which of the following statements is true in case of duality ?

a) Although the objective functions are different, their values (if exist) are same

b) Number of constraints in primal is same as number of constraints in dual

c) Dual of the dual is dual

d) From the primal simplex table, the solution of the dual variables cannot be obtained.

xi) Which of the following is *not* true for the feasible region ?

a) Is always unbounded for minimization problem

b) Contains the optimal solution

c) Can be found graphically

d) Is a convex set.

xii) In a single server Q-System, the service system is occupied 75% time and inter-arrival time is 4 minutes then, service time is

a) cannot be determined

b) 4 mins

c) 3 mins

d) 2 mins.

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GROUP – B

(Short Answer Type Questions)

Answer any *three* of the following. $3 \times 5 = 15$

2. Find the dual of the following problem :

Max $Z = x_1 - 2x_2 + 3x_3$

Subject to, $3x_1 + 2x_2 - x_3 = 8$

$2x_1 - x_2 + 5x_3 \geq 3$

$x_1 + 2x_2 - 3x_3 \leq 4$

$x_1, x_2 \geq 0$, x_3 is unrestricted in sign.

3. Find the initial solution to the following transportation problem using VAM :

		Destination				
		D_1	D_2	D_3	D_4	Supply
Factory	F_1	3	3	4	1	100
	F_2	4	2	4	2	125
	F_3	1	5	3	2	75
	Demand	120	80	75	25	300

4. For the following pay-off matrix for firm *A*, determine the optimal strategies for both the firms' and the value of the game :

Firm *B*

$$\text{Firm } A \begin{bmatrix} 3 & -1 & 4 & 6 & 7 \\ -1 & 8 & 2 & 4 & 12 \\ 16 & 8 & 6 & 14 & 12 \\ 1 & 11 & -4 & 2 & 1 \end{bmatrix}$$

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5. A self service store employs one cashier at its counter. 8 customers arrive on an average every 5 mins while the cashier can serve 10 customers in the same time. Assuming Poisson's distribution for arrival and exponential distribution for service rate determine :

   a) Average number of customers in the system

   b) Average number of customers in the queue

   c) Average time a customer spends in the system

   d) Average time a customer waits before being served.

6. With the help of a Quantity Cost curve, explain the significance of Economic Order Quantity. What are the limitations in using the formula for EOQ ?

GROUP – C

(Long Answer Type Questions)

Answer any *three* of the following. $3 \times 15 = 45$

7. a) Solve the following LPP by using simplex method :

   Max $Z = X_1 + 2X_2 + X_3$

   Subject to, $2X_1 + X_2 - X_3 \leq 2$

   $-2X_1 + X_2 - 5X_3 \geq -6$

   $4X_1 + X_2 + X_3 \leq 6;\ X_1, X_2, X_3 \geq 0$.

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   b) Solve the following Assignment Problem by Hungarian Method with the objective of incurring minimum cost :

	Job				
Mechanic	**1**	**2**	**3**	**4**	**5**
A	10	3	3	2	8
B	9	7	8	2	7
C	7	5	6	2	4
D	3	5	8	2	4
E	9	10	9	6	10

7 + 8

8. Arrivals at telephone booth are considered to be Poisson with an average time of 10 minutes between one arrival and the next. The length of the phone calls is assumed to be distributed exponentially, with a main of 3 minutes :

   a) What is the probability that a person arriving at the booth will have to wait ?

   b) The telephone department will install a second booth when convinced that an arrival would expect waiting for at least 3 minutes for a phone call. By how much should the flow of arrivals increase in order to justify a second booth ?

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c) What is the average length of the queue that forms from time to time ?

d) What is the probability that it will take a customer more than 10 minutes altogether to wait for the phone and complete his call ?

3 + 4 + 4 + 4

9. A small project is composed of 7 activities are listed in the table below. Activities are identified by their beginning (i) and end (j) node numbers :

Activity	Optimistic time (weeks)	Most likely (weeks)	Pessimistic time (weeks)
1-2	1	1	7
1-3	1	4	7
1-4	2	2	8
2-5	1	1	1
3-5	2	5	14
4-6	2	5	8
5-6	3	8	15

a) Draw the network diagram of the activities in the project.

b) Find the expected duration and variance of each activity.

c) What is expected project length ?

d) What is the probability that the project will be completed at least 4 weeks earlier than expected time ?

4 + 3 + 4 + 4

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10. a) Solve the following Linear Programming Problem using Charne's Big M Method :

Max $Z = 4X_1 + X_2$

Subject to $3X_1 + X_2 = 3$

$4X_1 + 3X_2 \geq 6$

$X_1 + 2X_2 \leq 4$

$X_1, X_2 \geq 0$

b) Following is the pay-off matrix for player A :

		B				
		I	II	III	IV	V
A	I	2	4	3	8	4
	II	5	6	3	7	8
	III	6	7	9	8	7
	IV	4	2	8	4	3

Using dominance property, obtain the optimal strategies for both the players and determine the value of the game. 9 + 6

11. a) A manufacturing company has determined from an analysis of its accounting and production data for a certain part, that its cost to purchase is Rs. 36 per order and Rs. 2 per part. Its inventory

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carrying charge is 9% of the average inventory. The demand of this part is 10,000 units per annum. Determine –

i) the economic order quantity.

ii) the optimal number of orders. 7

b) Write short notes on any *two* of the following : 2 × 4

i) Safety stock

ii) Floyed algorithm

iii) Convex set

iv) Saddle point.

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